Math 629 Spring 26 **Name: Mengxiang Jiang**

Turn this assignment in on Gradescope. All work must be neat and legible. Turn in a "final draft". There should be nothing scratched out and your work should flow generally from left to right and top to bottom.

1. Chapter 17 discusses Huygens's work on evolutes and involutes. Let the cycloid be given parametrically by

$$x = a(t - \sin t), \qquad y = a(1 - \cos t).$$

Show that

$$\frac{dy}{dx} = \frac{\sin t}{1 - \cos t} = \cot\left(\frac{t}{2}\right).$$

Then compute the element of arclength ds in terms of dt.

First, we compute dx and dy:

$$dx = a(1 - \cos t)dt, \qquad dy = a \sin t dt$$

So,

$$\frac{dy}{dx} = \frac{a \sin t dt}{a(1 - \cos t)dt} = \frac{\sin t}{1 - \cos t} = \cot\left(\frac{t}{2}\right).$$

For the element of arclength, we have

$$\begin{aligned} ds &= \sqrt{(dx)^2 + (dy)^2} \\ &= \sqrt{(a(1 - \cos t)dt)^2 + (a \sin t dt)^2} \\ &= a\sqrt{(1 - \cos t)^2 + \sin^2 t}dt \\ &= a\sqrt{1 - 2\cos t + \cos^2 t + \sin^2 t}dt \\ &= a\sqrt{2 - 2\cos t}dt \end{aligned}$$

2. Using modern techniques, put Johann Bernoulli's solution to the catenary problem, $dy = \frac{a}{\sqrt{x^2-a^2}}dx$ into a closed-form expression.

   The key identity that we will use is the following:

$$\frac{d}{dx}\cosh^{-1} x = \frac{1}{\sqrt{x^2-1}}.$$

   First, we can factor out a from the denominator:

$$dy = \frac{a}{\sqrt{x^2-a^2}}dx = \frac{a}{\sqrt{a^2(\frac{x^2}{a^2}-1)}}dx = \frac{1}{\sqrt{\frac{x^2}{a^2}-1}}dx.$$

   Now, we can make the substitution $u = \frac{x}{a}$, which gives us $du = \frac{1}{a}dx$ or $dx = a du$. Substituting this into the integral, we get:

$$\int dy = \int \frac{1}{\sqrt{u^2-1}} a du = a \int \frac{1}{\sqrt{u^2-1}} du.$$

   Using the identity, we can integrate:

$$y = a \int \frac{1}{\sqrt{u^2-1}} du = a\cosh^{-1} u + C = a\cosh^{-1}\left(\frac{x}{a}\right) + C.$$

3. Show why Lagrange's power series representation fails for the case $f(x) = e^{-1/x^2}$.

Suppose for contradiction that $f(x) = e^{-1/x^2}$ has a power series representation around $x = 0$. This means we can write:

$$f(x) = \sum_{n=0}^{\infty} a_n x^n,$$

where a_n are the coefficients of the power series. To find these coefficients, we can use the formula:

$$a_n = \frac{f^{(n)}(0)}{n!}.$$

Now, we need to compute the derivatives of $f(x)$ at $x = 0$. The first derivative is:

$$f'(x) = e^{-1/x^2} \cdot \frac{d}{dx}(-1/x^2) = e^{-1/x^2} \cdot \frac{2}{x^3}.$$

In general the n-th derivative of $f(x)$ will be of the form:

$$f^{(n)}(x) = P_n\left(\frac{1}{x}\right) e^{-1/x^2},$$

where $P_n(x)$ is some polynomial in x. As $x \to 0$, the term e^{-1/x^2} approaches zero faster than any polynomial can grow. Therefore, we have:

$$\lim_{x \to 0} f^{(n)}(x) = 0.$$

This means that all the coefficients a_n in the power series representation will be zero:

$$a_n = \frac{f^{(n)}(0)}{n!} = 0.$$

Thus, the power series representation of $f(x)$ around $x = 0$ is

$$f(x) = \sum_{n=0}^{\infty} 0 \cdot x^n = 0.$$

However, this is a contradiction because $f(x) = e^{-1/x^2}$ is not identically zero for all x.